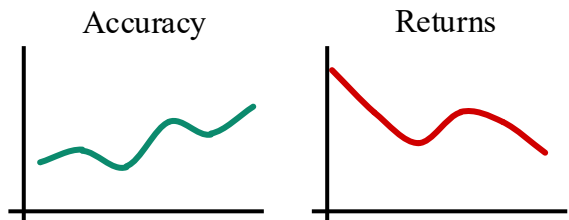
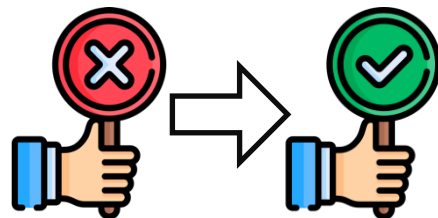


Direction-label target



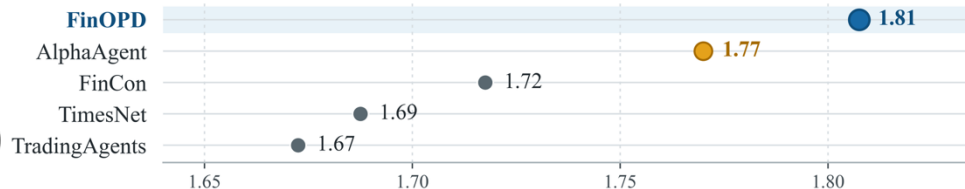
Accuracy can rise while returns fall.



Hindsight-PnL Supervision

Portfolio Sharpe $\uparrow$

Top 5 of 11 methods · 2025 post-cutoff · 4-asset equal-weight · same cost-aware protocol



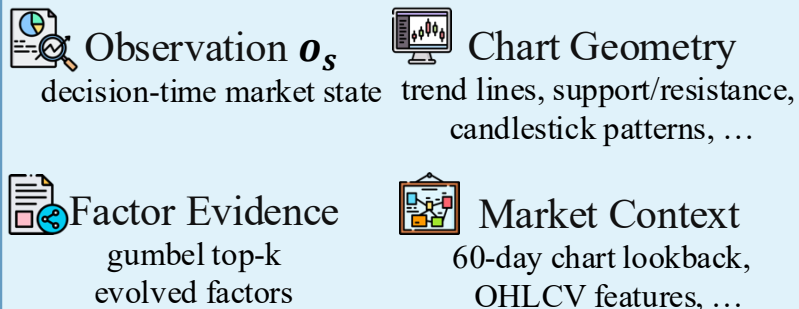
FinOPD attains the highest portfolio among compared methods.

Hindsight-PnL OPD Mechanism



STUDENT $p_S(\theta)$
decision-time evidence

Input:



Output: uncertain policy. $p_S(y, | \mathbf{o}_t y_{<n})$



p_S tokens

Token-level JSD

$$D_\beta(p_T \parallel p_S) = \beta \cdot KL(p_T \parallel m) + (1 - \beta) \cdot KL(p_S \parallel m)$$
$$m = \beta p_T + (1 - \beta) p_S$$

Alignment Guard

1. $KL(p_S \parallel p_{\text{ref}}) < \epsilon$, $\epsilon = 0.1$
2. per-token KL clipping: $c = 5.0$

LoRA $_{\theta}$ Update

update student LoRA_{θ} ;
keep teacher θ_0 frozen.

Agentic Evolution

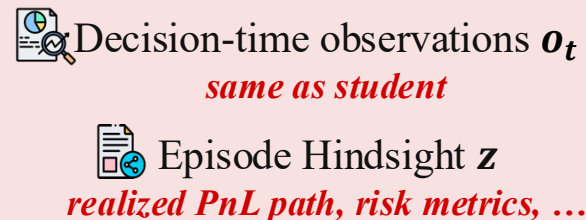
update memory $\mathbf{m}$;
update agents credit $\mathbf{r}$

Next rollout window



Frozen TEACHER $p_T(\theta_0)$
post-horizon query

Input:



Output:

profit-informed policy. $p_T(y, | \mathbf{o}_t y_{<n} \hat{y})$

p_T tokens

